Causal Attitude Networks on the Substrate

What Moved, What Did Not, and What the Mean Field Costs

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DRAFT — draft-2026-09-06.1r4 (6 September 2026). Circulated for comment, not as a finished result. **This revision reports the programme's record to 14 August 2026**; its first revision reported the same record as it stood on the morning of 13 August, and the record moved past all three of that revision's findings within two days. Every remark below therefore carries the date of the rung it reports. Two of the first revision's five falsifiability commitments have fired against the record; both are reported as fired. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

The Causal Attitude Network (Dalege et al., 2016) models an attitude as an Ising-family network of interacting evaluative components. *The current system is a native-default discrete graph-rule implementation*: its ordinary transition, learning, evidence-decay, and identity-update paths run as scoped rules over the target microtheory. A prior hosted mean-field self-consistency solver is retained in this paper as a dated implementation and experimental record, not as the current ordinary backend. The migration did not require a new domain-specific term kind, but it did require the compiled runtime to expose scoped rule execution; the earlier claim that no compiled-runtime capability changed was therefore too strong. The former per-network scope blocker is resolved: native settling now isolates the target microtheory, while the deliberately retained unscoped runner remains graph-global.

The native transition commits -1, 0, or $+1$ from a temperature-scaled local field, and is asynchronous because each rule commitment is visible to the next match. A finite-state convergence argument is available for symmetric couplings, fixed nonnegative thresholds, no self-coupling, and asynchronous updates with the stated tie rule. This includes the ternary state space. Deployment under learning, decay, changing temperature, or asymmetric couplings remains an operational question; the fixed-parameter proof does not cover those paths. The historical synchronous mean-field cycle and its damping correction, the registered resolution results, and the performance figures below are reported as dated measurements rather than claims about the present primary backend. Loopy belief propagation and bounded-treewidth junction-tree inference are available as explicit, Python-orchestrated auxiliary tools: neither is an installed graph-rule mechanism, automatic dispatch, nor a general convergence or tractability guarantee. All findings are reported as measured rather than reinterpreted.

1 What was moved, what a move would mean, and on what standard

An attitude in the CAN model is a network of evaluative components with pairwise couplings and external fields, and its behaviour is the behaviour of an Ising system: relaxation toward

low-energy configurations, frustration where couplings conflict, and a temperature-like parameter governing how sharply it settles.

Moving a formalism onto this substrate means more than making it run there. The substrate's claim — the substrate is the one set out in *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation) — is that decisions are taken by graph rules over graph data; a model that computes its result in a compiled routine and hands back an answer has been *hosted*, not moved. The distinction is the whole of the alignment question below, and it is not rhetorical: the two cases differ in what can subsequently be inspected, mutated, minted and reasoned over by the rest of the account. It is also the criterion that decides Remark 2.2, and the criterion was met after this paper's first revision said it had not been.

The standard, stated once. Every figure below is a *measurement* reported from this programme's own record rather than from a published source. It is consequently *parity-on-credit*: the figures receive no greater evidential standing than the reader grants the unavailable record. What is offered is a pre-registered implementation record which is not circulated with these papers, and it is a claim a reader may decline to credit on exactly the terms Graphic Equalisation sets for the same case. Each rung of that record fixes its own commit, seed and sample before it runs, and reports what was compared, what was found, and the scope the run holds under; it names no implementation identifiers, and neither does this paper. The standard is not re-disclaimed at each remark. What each remark does carry is the *date* of the rung it reports, because the defect this revision repairs was a dating one.

2 Alignment: reimplemented twice

The model arrived as a hosted solver: a compiled mean-field self-consistency routine, exposed to the host and driven by seed-resident network data. The rules written against it did nothing, and the reason was worth having.

Remark 2.1 (The hosted stage, and the null that dated within a day). *(Record of 13 August 2026.)* There was no term surface for the dynamics at all, so the fifteen rules written against the network were inert — and the obvious diagnosis, missing or malformed match clauses, was not the reason. The relaxation existed only across a host-language boundary, so a rule could not express it, could not be matched against it, and could not mint against it. The rules were not broken; there was nothing for them to be written in terms of. Exactly one of the fifteen described the *recognition* half rather than the update half, and that one was wired with real match, where and effects clauses using only the existing term vocabulary; it works on the production path. *So the boundary was not a wall*, and the narrow statement that followed — what had moved was the network's data and its results, not its dynamics — was true for about one day.

Remark 2.2 (The dynamics are native, and two engine findings came with them). *(Record of 13 August 2026.)* The update was expressed in the existing term vocabulary as five rules and four admissibility productions, with *no new domain-specific term kind*; the subsequent native-default deployment also added a compiled runner capability for scoped rule execution. A concept's local field is the sum of its external and identity fields and the coupling-weighted states of its outgoing neighbours; a rule commits the thresholded value of that field — three-valued, with a neutral band — whenever it differs from the held state. A physical symmetric coupling must consequently be represented by two directed causal-influence edges: the implementation does not infer an undirected coupling. The update is asynchronous by the engine's own execution model rather than by arrangement: one progressing binding is committed per pass, and the next match reads the graph as it now stands. Energy descent applies only if paired weights are equal, self-couplings are zero, fields and thresholds are fixed during settling, and updates are asynchronous. The implementation must enforce or check those conditions; two directed edges

alone do not establish symmetry. The finite three-state case under those conditions is derived below; the draw record does not establish them for arbitrary changing networks. This meets the criterion Section 1 gives for *moved* rather than *hosted*.

Two engine-level findings were made and corrected while building, and both belong to the machinery rather than to the model. *First*: where sibling rules compete to update the same kind of node, the pass scheduler lets the first with a replenished candidate pool take every pass and starve the others indefinitely. It presented as non-convergence and was diagnosed as its opposite — nodes frozen at their initial value, the system already at a reachable fixed point it could not reach — and was fixed by collapsing the sign branches into one rule that computes its target. *Second*: the decoder's search prefers a redundant composite explanation to a clean single exact match at equal coverage, because a coverage tie is broken on accumulated fit rather than on fewest parts. That is an Occam violation in a tie-break; it was worked around by scoping the codebook to one candidate at a time, and is reported rather than fixed.

For a fixed positive inverse temperature, absorb its scaling into J and h (or divide the threshold by it). For the fixed symmetric three-state case, put $s_i \in \{-1, 0, 1\}$, $J_{ij} = J_{ji}$, $J_{ii} = 0$ and $\theta_i \geq 0$. The local field is $H_i = h_i + \sum_j J_{ij} s_j$, and the update selects $+1$ above θ_i, -1 below $-\theta_i$, and zero otherwise. Define

$$E(s) = -\frac{1}{2} \sum_{i,j} J_{ij} s_i s_j - \sum_i h_i s_i + \sum_i \theta_i s_i^2.$$

Changing only s_i changes E by $-H_i(s_i' - s_i) + \theta_i((s_i')^2 - s_i^2)$. The threshold rule minimises this coordinate energy. Each change strictly lowers E, except at threshold equality when choosing zero leaves E equal and lowers $\sum_i s_i^2$. Thus the pair $(E, \sum_i s_i^2)$ decreases lexicographically on a finite state space. A fair asynchronous schedule terminates at a fixed point. Learning or changing fields during settling, asymmetry, and synchronous multi-node changes are outside this proof.

Current execution boundary. The former per-network scope blocker is resolved. Native consumers call scoped settling, which limits rule firing to the target microtheory while preserving the graph-rule guards for state updates, Hebbian learning, evidence decay, and identity updates. An unscoped runner remains available and deliberately graph-global; it advances unrelated microtheories and is therefore not used by ordinary per-network lifecycle paths. A microtheory without an explicit non-discrete opt-out uses the native three-state path by default; the retained continuous route is an explicit, non-production opt-out rather than a coequal ordinary backend.

"Full native" names the graph-rule transition and plasticity mechanism, not every computation that can be composed around it. Counterfactual prediction and multi-start orchestration are bounded host-language procedures over the native local-field rule; graph-resident candidates and an argmin select the retained multi-start result. Basin recognition classifies a settled state against held signatures; it does not enumerate basins. Likewise, a concept shared by two microtheories has one committed graph state, not independent per-microtheory states; restricted-network hypothetical state is an auxiliary calculation when that distinction is needed. These are explicit boundaries, rather than reasons to misdescribe ordinary CAN dynamics as hosted.

3 The solver: what the oscillation turned out to be

Remark 3.1 (The oscillation was the update schedule, and the record found it). *(Record of 13 August 2026.)* What was observed is not in doubt and reproduces exactly. On a fixed eight-node draw with a fixed nonzero field per node and a fixed start, the relaxation returns different states at 400 and 401 iterations (opposite-parity gap 5.615746) and the same state at 400 and 402: a persistent period-2 cycle against a same-parity control. This paper's first revision reported that

as a genuine binary between cohering and failing to cohere — an operation that either resolves or returns a determinate nothing — already present, unexploited, in the borrowed model's own mathematics, and so as an instance of the claim that such failure comes in one manner and not in kinds (Graphic Equalisation, Proposition 9.1).

It was the update schedule. The relaxation iterated $\sigma_i \leftarrow \tanh(\beta(\sum_j J_{ij}\sigma_j + h_i))$ fully synchronously — every node's update in a sweep reading the previous sweep's state — which is the textbook source of period-2 limit cycles in this family; a second, byte-identical copy of the same iteration was found in a sibling routine that the theoretical audit preceding the fix had not named. Blending each freshly computed value with its own prior value, at a factor registered in advance rather than tuned against these networks, makes the same draw converge cleanly (gap 0.000000); setting that factor to 1 reproduces the old numbers bit for bit on the same binary. The before and the after are one draw and one code change apart, not two experiments. Blending cannot move or remove a fixed point, only the trajectory to one, which is why nothing that already converged changed — and why the damped fixed point is a nearby genuine one rather than the average of the old cycle's two phases, exactly as a frustrated network with several fixed points should give.

The elimination is total across everything re-tested: 126 cells — the original 24-trial registered sweep, previously 6/12 and 8/12 oscillating by condition; a 30-cell sharpening previously up to 100% oscillating at $N = 32$; and a 72-cell extension past the registered scope to $N = 64$ and $\beta = 4.0$ — converge cleanly, every one, gap exactly zero. So the correction is precise, and it is the useful part of the whole arc: *the binary this account has elsewhere was not present in the borrowed mathematics; what was present was a Jacobi iteration.* What is *not* established is the general claim. No scheduling-independent failure to converge was found anywhere this search looked, and that is a bounded negative — complete graphs only, $\beta \leq 4.0$, $N \leq 64$, one initial condition per trial — rather than a proof that none exists. One later rung of the arc still calls the binary real and confirmed again; it ran on the uncorrected solver, and re-running its sweep under the corrected one is named in the record as a follow-on it did not perform.

Frustration convention. The state-map scorer uses $F(s) = \sum_{(i,j)}(|J_{ij}| - J_{ij}s_is_j)$ over its enumerated couplings. A satisfied binary edge contributes zero, an opposed edge $2|J_{ij}|$, and an edge incident to a neutral state $|J_{ij}|$. The neutral option can reduce this score by avoiding opposed commitments without showing uncertainty has been resolved. Vocabulary comparisons require identical couplings and scoring conventions; this is a metric-specific result, not a theorem about attitude quality.

4 Performance: the complexity question is the wrong one

The natural objection to putting an Ising model anywhere is that its ground-state problem is hard. Barahona (1982) makes the sharp version available: the planar two-dimensional problem *without* an external field is solvable in polynomial time, while the same planar problem *with* a field, and the three-dimensional problem either way, is NP-hard. Every network here carries a nonzero field, so the hard case is the relevant one.

And it does not bind the ordinary CAN transition, because that transition is a discrete, local graph-rule relaxation rather than an attempt to find every exact ground state. The historical hosted solver computed a mean-field self-consistency fixed point; its performance and convergence results remain below as an experimental record, not as a description of the current default backend. Exact inference is an optional tool selected only after a treewidth gate, and refuses rather than attempting the exponential case. The practical question is therefore which explicitly selected procedure is appropriate for a stated network, objective, and resource bound.

Remark 4.1 (The solver ladder is an auxiliary toolbox). *(Record of 14 August 2026.)* Mean field is the first and weakest step of a known ladder — mean field, then Bethe and loopy belief

propagation, then Kikuchi and generalized belief propagation (Yedidia et al., 2005) — and it is weakest *precisely* in the regime of loops, strong local correlation and frustration, which is the regime the frustration and basin audits exist to detect. The following results concern explicit Python-orchestrated solver tools over a graph snapshot; they do not make loopy propagation or clique-tree inference installed native graph rules, automatic dispatch, or a general guarantee of convergence or tractability. They are retained as dated experimental records of those tools, while the graph-rule CAN transition remains discrete and native.

Loopy propagation, both variants. Max-product is the theoretically appropriate one here, because everything else in this toolbox is organised around a joint configuration rather than a per-node marginal; on the dense frustrated case it is never worse than sum-product across eight draws, and on every draw where it converges it ties or beats a twenty-restart search. Its convergence reliability then *inverts across topology*: it converges on 5 of 8 dense adversarial draws but only 4 of 15 real sparse ones — the opposite of the naive expectation that sparser is easier — and the cause was measured rather than left as an anomaly, a sustained period-4 message oscillation traced to one structural 3-cycle plus small-integer weights producing exact ties between candidate scores. Sum-product converges everywhere (15 of 15) and to worse joint states (mean frustration 2.142 against 1.693 for the restart search): the textbook warning that a marginal argmax is not a good joint configuration, measured here rather than cited. Neither variant reliably beats the restart search on the deployed networks, and both cost 20–200× its wall-clock per solve. So the caveat held where it was aimed, and was not the whole story: the worse convergence is on the sparse networks, where it predicted none.

Exact inference, where the width permits. Off that ladder rather than on it, an explicit Python junction-tree MAP tool provides exact minimum-energy inference for low-treewidth networks. It obtains a graph snapshot, refuses before exponential work when the configured treewidth bound is exceeded, and may commit the selected exact discrete state when invoked to do so; it is neither an automatic fallback nor a graph-rule transition. The recorded comparisons agree with literal brute-force enumeration on five real networks, float-tight, and cross-check against the energy function every other consumer uses. On a real cycle-topology network the recorded run finds a strictly lower-energy state (-5.743 against -5.102) than twenty random restarts. That last is the concrete content of what "exact" buys in its admissible range: a widened search finds the best of the basins its restarts land in, and this procedure instead solves the bounded-width objective directly.

Tractability, measured. The restart search exists because local-minimum risk was measured first, and not only on the adversarial case: 14 of 30 real deployed networks (46.7%) show nonzero spread across twenty starts, so a sparse near-tree topology does not by itself guarantee a benign landscape. Treewidth was measured exactly on both graphs — 2 for the real template, 15 for the dense stress case — which is what decides where exact inference applies at all. And the real coupling matrix has effective rank ≈ 1.98 of 5: the couplings this model carries are substantially less independent than their count suggests.

5 Results

Remark 5.1 (The registered scaling prediction is reversed, and its magnitudes moved). *(Record of 13 August 2026.)* The primary registered prediction — that coarse quantization would yield fewer distinct basins than fine, with the gap widening in network size — *is not confirmed, and the measured direction is closer to reversed.* As first measured, coarse yields *more* distinct basins (6 against 4 at $N = 4$; 15 against 4 at $N = 8$), a higher entropy-weighted effective basin count, and higher mean frustration (2.10 against 1.55 at $N = 4$; 15.32 against 11.06 at $N = 8$); at $N = 16$ the two conditions converge to near-parity on every measure. The manipulation check holds throughout — coarse quantization does produce larger per-edge representational error at every N — so the intervention worked and the prediction about its consequence was wrong.

That is the useful shape of a null and it is reported here rather than reinterpreted into a different prediction that would have survived.

Those figures were measured on the uncorrected solver of Remark 3.1. The reversal was subsequently replicated on five independent coupling draws per cell and the replicated cells re-run under the corrected one. *The direction holds*: the effective basin count is higher for coarse at both sizes (2.358 against 2.145 at $N = 4$; 3.616 against 2.088 at $N = 8$). *The magnitudes do not*: frustration falls by roughly 7–10× at $N = 4$ and 2–2.5× at $N = 8$, and the effective basin count roughly halves, because restarts landing on different phases of an oscillation had been counted as distinct basins. And one sub-claim does not survive at all — the concentration reading, that a coarse landscape is the less concentrated of the two, is gone at $N = 4$ (0.565 against 0.560) and reversed at $N = 8$ (0.405 against 0.575), which retires the gloss that coarse belief is the more easily knocked between alternatives. The single-draw frustration figures of the first paragraph were not themselves re-measured; the re-check ran on the replicated cells.

Two further results the arc produced, both on the uncorrected solver and both carrying that caveat. *A null*: there is no crossover network size anywhere in $\{4, 8, 12, 16, 24, 32\}$ under a criterion registered in advance, on either the plain or the structured coupling design. The criterion never fires, and the pattern does not approach it — the sign of the effective-basin gap flips three times across the six sizes. *And the strongest positive result in the whole arc*, which is not about bin count at all: the sharpest resolution effect measured anywhere is the presence or absence of a *neutral option*. A two-value vocabulary produces 1.6–2× the frustration of every vocabulary with three or more values, at every size tested (22.6 against a 13.9–15.3 band at $N = 8$; 122.9 against 48.4–61.9 at $N = 16$; 501.3 against 246.9–275.2 at $N = 32$), while three through eleven values show no reliable ordering among themselves. Under the corrected solver this is the one earlier finding that does not merely survive but *sharpens*, to 2.36× at $N = 8$ and 2.71× at $N = 16$. Removing the neutral option is not coarsening a scale; it is removing the representation's own way of declining to commit, and that is what the frustration is measuring.

Remark 5.2 (Resolution separates the basins, with a control that says so). *(Record of 13 August 2026.)* A second registered run, on a population of simulated residents each carrying its own network at its own representational resolution, confirmed both of its predictions.

The manipulation check is large: the coarse resident's mean rounding error is about 9.2 times the fine resident's (0.553 against 0.060). The primary prediction — deliberately not the naive basin-count intuition, that having already been falsified above on the same engine — is that under *identical* scenario evidence the coarse network relaxes to a positive-energy, highly frustrated basin (frustration 9.86, energy +5.19) while the fine one relaxes to a negative-energy, low-frustration basin (frustration 1.62, energy −1.77). It does.

A population-wide check across one hundred residents, reported as exploratory rather than pre-registered to the same standard, shows the same direction on average (mean frustration 3.29 in the low-resolution group against 1.38 in the high). *The control is what makes this worth reporting*: holding resolution fixed across the population collapses the trait-to-frustration correlation from −0.33 to −0.08, so resolution rather than some correlated trait is carrying it.

Reproduction boundary. The numerical claims below retain their original run dates and scope. A reader needs the referenced run artifacts, configuration and code revision to reproduce them; the paper alone is not a complete reproduction package. A record-dependent commitment is not independently runnable until those artifacts are supplied. No new empirical run is reported in this revision.

6 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a

mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

Two of the five commitments made in this paper's first revision have *fired* against the record this paper reports from. Both are recorded as fired and replaced below, not quietly removed.

1. [SHAPE] **Ordinary CAN dynamics are native-default and scoped.** Fails if settling a target microtheory advances an unrelated microtheory, if a microtheory without an explicit non-discrete opt-out does not take the native three-state path, or if the graph-rule transition diverges from its specified local-field threshold under an auditable fixture. The relevant comparison is no longer against the historical hosted mean-field solver, because that solver is not the normal supported backend. The first revision's partial-migration commitment fired: the dynamics do have a term surface. The earlier scope-boundary version of this commitment has also fired, because scoped execution now exists; its replacement concerns isolation and default dispatch rather than merely the possibility of expressing the rules.

2. [DETAIL] **Resolution drives basin quality.** Fails if the frustration and energy separation survives holding resolution fixed, or if the population correlation does not collapse under that control.

3. [DETAIL] **The scaling prediction is reversed and not merely unconfirmed.** Fails if the basin counts recover the registered direction at $N = 4$ and $N = 8$ under a larger sample *on the corrected solver*. The first revision's form of this commitment — a larger sample on the uncorrected solver — has been run and did not fire: five independent draws per cell hold the direction. It is restated here against the corrected solver, on which the direction also holds (Remark 5.1) and the magnitudes do not.

4. [SHAPE] **Exact inference does not change what the separation says.** Fails if the frustration and energy separation of Remark 5.2 does not survive exact clique-tree inference on the treewidth-2 networks where it applies. The first revision's form of this — "fails if an exact solver at these network sizes changes the reported quantities materially" — had no instrument to fire against. It now has one (Remark 4.1), and running it against Remark 5.2 is the one thing on this list not yet done.

5. [DETAIL] **The oscillation is the solver's and not the model's.** Fails if a damped, asynchronous, or independently written solver reproduces a period-2 cycle on any of the 126 re-tested cells. *This replaces a commitment that fired.* The first revision committed, as [SHAPE], that "the oscillation is the model's and not the implementation's — fails if the two-cycle disappears under an equivalent solver written independently." It disappears under damping, on every cell re-tested (Remark 3.1). The replacement is [DETAIL] because what is now claimed is a fact about one iteration scheme, not a shape the account requires.

6. [DETAIL] **Max-product's convergence deficit on the sparse networks is the mechanism named.** Fails if breaking the tied candidate scores — perturbing the small-integer weights — does not restore convergence on the networks where the period-4 message oscillation was measured.

References

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